

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

First Semester B.Tech Degree Regular and Supplementary Examination December 2022 (2019 Scheme)

Course Code: MAT 101**Course Name: LINEAR ALGEBRA AND CALCULUS
(2019 -Scheme)**

Max. Marks: 100

Duration: 3 Hours

PART A*Answer all questions, each carries 3 marks*

Marks

- 1 Find the rank of the matrix $\begin{bmatrix} 3 & 0 & 2 & 2 \\ -6 & 42 & 24 & 54 \\ 21 & -21 & 0 & -15 \end{bmatrix}$ (3)
- 2 Show that the quadratic form $3x_1^2 + 22x_1x_2 + 3x_2^2 = 0$ represents a pair of straight lines by transforming it into principal axes form. (3)
- 3 Find the slope of the surface $z = \sqrt{x^2 + 4y^2}$ in the x-direction at the point (1, 2) (3)
- 4 Prove that $Z_{xy} = Z_{yx}$ where $Z = x^2y + 5y^3$ (3)
- 5 Evaluate $\int_1^a \int_1^b x^2 y \, dx \, dy$. (3)
- 6 Use double integral to find the area of the plane region enclosed by the curves $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \frac{\pi}{4}$. (3)
- 7 Determine whether the series $\sum_{k=1}^{\infty} \frac{4^{k+2}}{7^{k-1}}$ converges and if so find its sum (3)
- 8 Examine the convergence of $\sum_{k=1}^{\infty} \frac{1}{k!}$ (3)
- 9 Find the Taylor series expansion of $f(x) = \cos x$ about $x = \pi$ up to third degree terms. (3)
- 10 Find the half range sine series expansion of $f(x) = \pi x - x^2$ in $(0, \pi)$ (3)

PART B*Answer one full question from each module, each question carries 14 marks.***MODULE 1**

- 11 a Solve the following linear system of equations using Gauss elimination method (7)
- $$\begin{aligned} x + y + z &= 2 \\ y + z &= -2 \\ 4y + 6z &= -12 \end{aligned}$$
- b Find the eigenvalues and eigenvectors of $\begin{bmatrix} 1 & 1 & 2 \\ -1 & 2 & 1 \\ 0 & 1 & 3 \end{bmatrix}$ (7)

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- 12 a Solve the following linear system of equations using Gauss elimination method. (7)
- $$\begin{aligned} 2x - 3y - 3z + 6w &= 2 \\ y + z - 2w &= 0 \\ 4x + y + z - 2w &= 4 \end{aligned}$$

- b Find the matrix of transformation that diagonalize the matrix $A = \begin{bmatrix} -19 & 7 \\ -42 & 16 \end{bmatrix}$. (7)
- Also write the diagonal matrix.

MODULE 2

- 13 a If $w = f(P, Q, R)$ where $P = \frac{x}{y}$, $Q = \frac{y}{z}$, $R = \frac{z}{x}$ prove that (7)
- $$x \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} + z \frac{\partial w}{\partial z} = 0.$$

- b Locate all relative extrema and saddle points of $f(x, y) = 2xy - x^3 - y^2$. (7)

- 14 a The length and width of a rectangle are measured with errors of at most $r\%$, (7)
- where r is small. Use differentials to approximate the maximum percentage error in the calculated length of the diagonal.

- b Find the local linear approximation L of $f(x, y) = \sqrt{x^2 + y^2}$ at the point (7)
- $P(3, 4)$. Compute the error in approximation f by L at the point $Q(3.04, 3.98)$.

MODULE 3

- 15 a Evaluate $\iint_R \frac{\sin y}{y} dA$, where R is the triangular region bounded by the y -axis, $y = x$ (7)
- and $y = \pi$.

- b Evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} dx dy$ by reversing the order of integration. (7)

- 16 a By converting into polar coordinates evaluate $\int_{-1}^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)^{\frac{3}{2}} dy dx$. (7)

- b Find the volume of the solid in the first octant bounded by the coordinate planes and the plane $x + y + z = 1$. (7)

MODULE 4

- 17 a Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{k(k+1)}{(k+3)(k+2)(k+5)}$ (ii) $\sum_{k=1}^{\infty} k^{50} e^{-k}$ (7)

- b Test the convergence of the series $1 + \frac{1.3}{3!} + \frac{1.3.5}{5!} + \frac{1.3.5.7}{7!} + \dots$ (7)

- 18 a Determine whether the alternating series $\sum_{k=1}^{\infty} (-1)^{k+1} \frac{k+7}{k(k+4)}$ is conditionally (7)
- convergent or not.

- b Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{[\pi(k+1)]^k}{k^{k+1}}$ (ii) $\sum_{k=1}^{\infty} \frac{1}{(2k+5)^{17}}$ (7)

MODULE 5

- 19 a Find the Fourier series expansion of $f(x) = \frac{1}{2}(\pi - x)$ in $0 < x < 2\pi$ (7)
- b Find the half range Fourier sine and cosine series of $f(x) = x$ in $(0, \pi)$ (7)
- 20 a Find the Fourier series expansion of $f(x) = \begin{cases} \frac{\pi}{2} + x & \text{for } -\pi \leq x \leq 0 \\ \frac{\pi}{2} - x & \text{for } 0 \leq x \leq \pi \end{cases}$. Hence (7)
- show that $1 + \frac{1}{3^2} + \frac{1}{5^2} + \frac{1}{7^2} + \dots = \frac{\pi^2}{8}$
- b Obtain the half range Fourier sine series of $f(x) = \begin{cases} x, & 0 < x < 1 \\ 2 - x, & 1 < x < 2 \end{cases}$ (7)
